

Cambrian explosion

About 530 million years ago, a huge range of new animals appeared in the seas, including the first creatures with clear legs, heads, sense organs, skeletons, and shells. All the main categories of invertebrates (animals without backbones) known today seem to have evolved almost at once, as well as some weird creatures quite unlike anything else. Scientists call this mysterious burst of life the Cambrian explosion.

Anomalocaris

a-NOM-a-low-CAR-iss

- **When** 505 million years ago (Middle Cambrian)
- **Fossil location** Canada, S. China
- **Habitat** Oceans
- **Length** Up to 3 ft (1 m)

A little like a giant shrimp, *Anomalocaris* (below) was the largest animal found in the Burgess Shale fossil bed of Canada (see box). Experts suspect it was the top predator in Cambrian seas and used a pair of spiked claws attached to its head to grasp prey such as trilobites. It had no legs but could swim by flexing its segmented body and waving the flaps on its sides. Large compound eyes indicate it had good vision and hunted by sight.

Wiwaxia

we-WAX-ee-a

- **When** 505 million years ago (Middle Cambrian)
- **Fossil location** Canada
- **Habitat** Seafloor
- **Length** 1–2 in (3–5 cm)

Wiwaxia looked like a tiny, armored porcupine, since its body was covered with protective spines and rows of overlapping armor plates. Its flat lower surface, where the mouth was located, did not have any protection. The mouth had two or three rows of sharp, conical teeth that may have been used to scrape algae from the seabed. *Wiwaxia* had no distinct head or tail and was probably blind, relying on touch and smell to find the way.



▲ THIS *Wiwaxia* fossil from the Burgess Shale fossil bed in Canada is about 500 million years old. The armorlike plates on its back are called sclerites.



▲ FOSSILS of *Anomalocaris* often show only a small part of the body. This is one of its feeding claws.